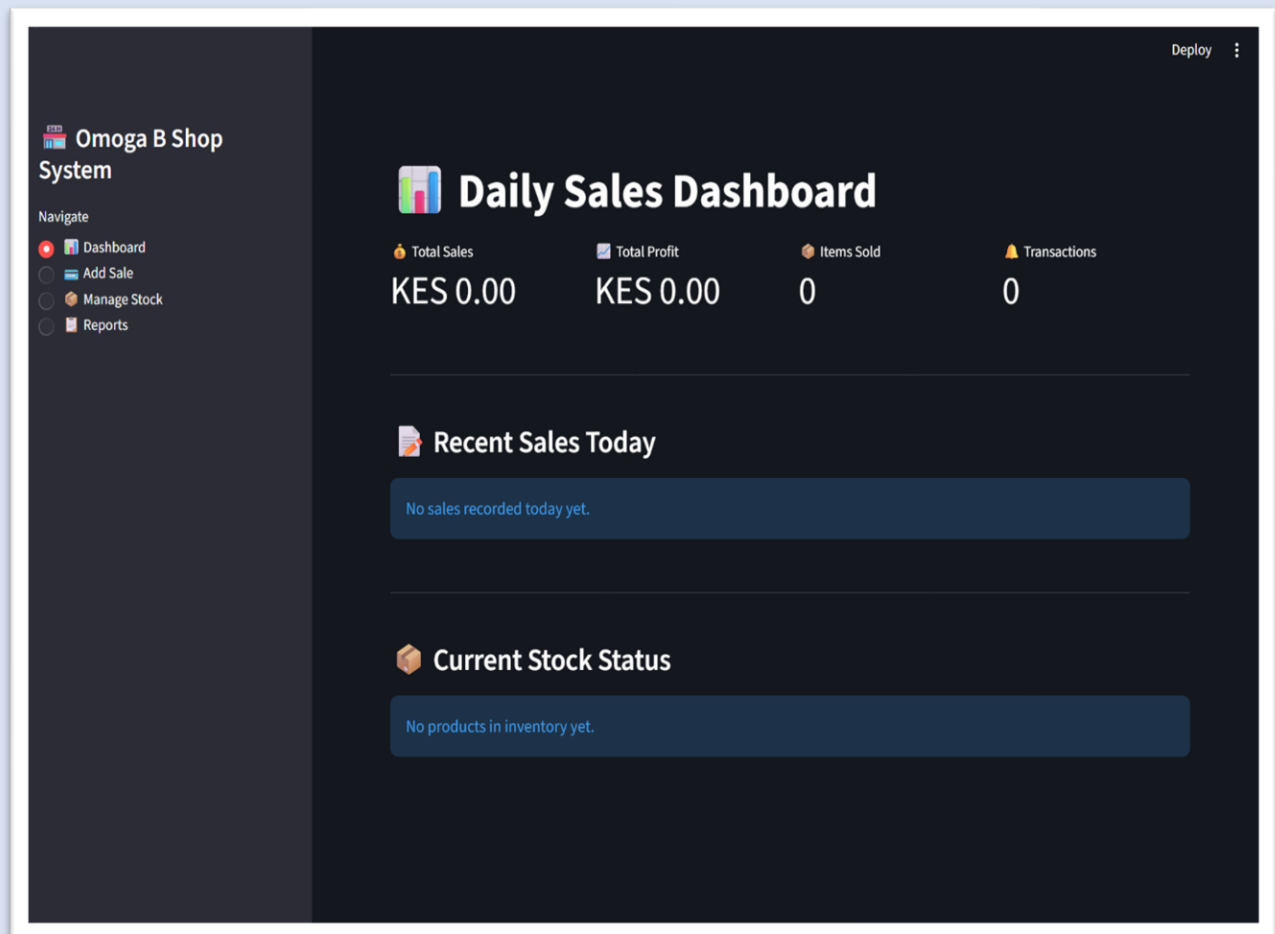




THE EXECUTIVE SUMMARY AND USER INTERFACE (DASHBOARD)





EXECUTIVE SUMMARY

Retail Sales & Profit Management System

The **Retail Sales & Profit Management System** is a real-time sales and inventory management solution developed to help small businesses efficiently monitor daily operations, track profitability, and manage stock levels. The system addresses common challenges faced by retail shops, such as manual record-keeping, inaccurate stock tracking, and limited visibility into business performance.

Built using **Python, Streamlit, SQLite, and Pandas**, the application provides an interactive dashboard that displays key business metrics, including total sales, profit generated, items sold, and transaction counts. It enables users to record sales transactions, automatically update inventory levels, manage product information, and generate daily, weekly, and custom date-range reports for business analysis.

The system incorporates a shopping cart-based sales process, supports multiple payment methods, and performs automatic profit calculations for each transaction. Through its inventory management module, users can add, update, and monitor products while maintaining accurate stock records. The reporting component provides valuable insights into sales trends and profitability, supporting data-driven decision-making.

Overall, the Retail Sales & Profit Management System demonstrates the practical application of software development, database management, business intelligence, and data analytics techniques to create an affordable and user-friendly solution for small retail businesses seeking to improve operational efficiency and financial performance.

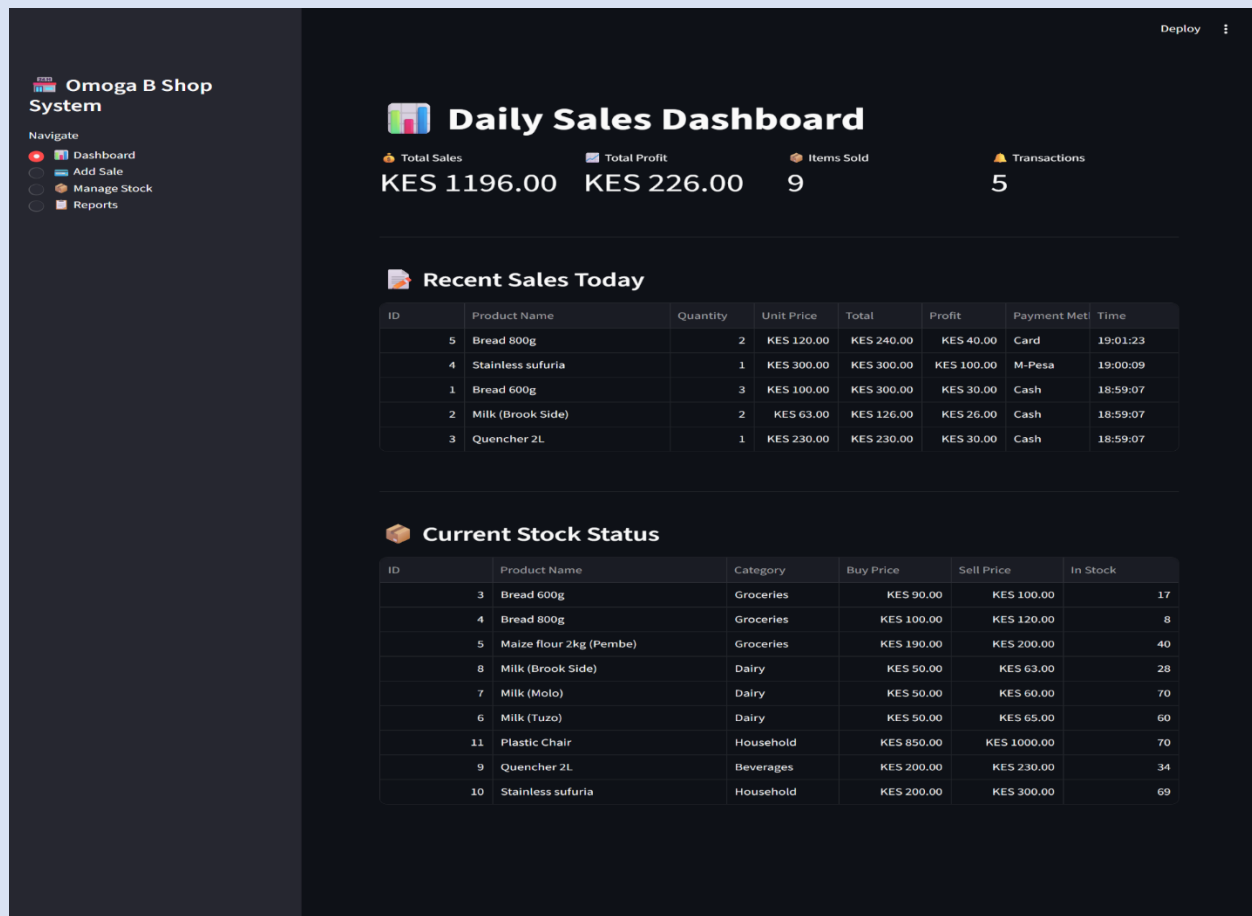


USER INTERFACE

A. Dashboard Tab - Overview & Functionality

The **Dashboard** is the central control panel of the system, designed to give a real-time snapshot of business performance.

It functions as a **summary and monitoring interface**, where key metrics and operational insights are displayed at a glance.



Key Functions in the Dashboard Tab

1. Sales Summary Metrics

At the top section, the dashboard displays high-level KPIs:

- **Total Sales** – Overall revenue generated for the selected period
- **Total Profit** – Net earnings after costs
- **Items Sold** – Total quantity of products sold
- **Transactions** – Number of completed sales operations



This section provides instant business performance insight without needing to open reports.

2. Recent Sales View

This section shows the latest transactions in a structured table:

- Product name
- Quantity sold
- Unit price
- Total value
- Profit per item
- Payment method (Cash, M-Pesa, Card)
- Time of sale

Purpose: Enables quick tracking of daily activity and verification of sales entries.

3. Current Stock Status

This section monitors inventory levels in real time:

- Product details (name, category)
- Buying vs selling price
- Stock remaining

Purpose:

- Prevent stock-outs
- Support restocking decisions
- Track product performance

Overall UI Role of Dashboard Tab

The dashboard acts as:

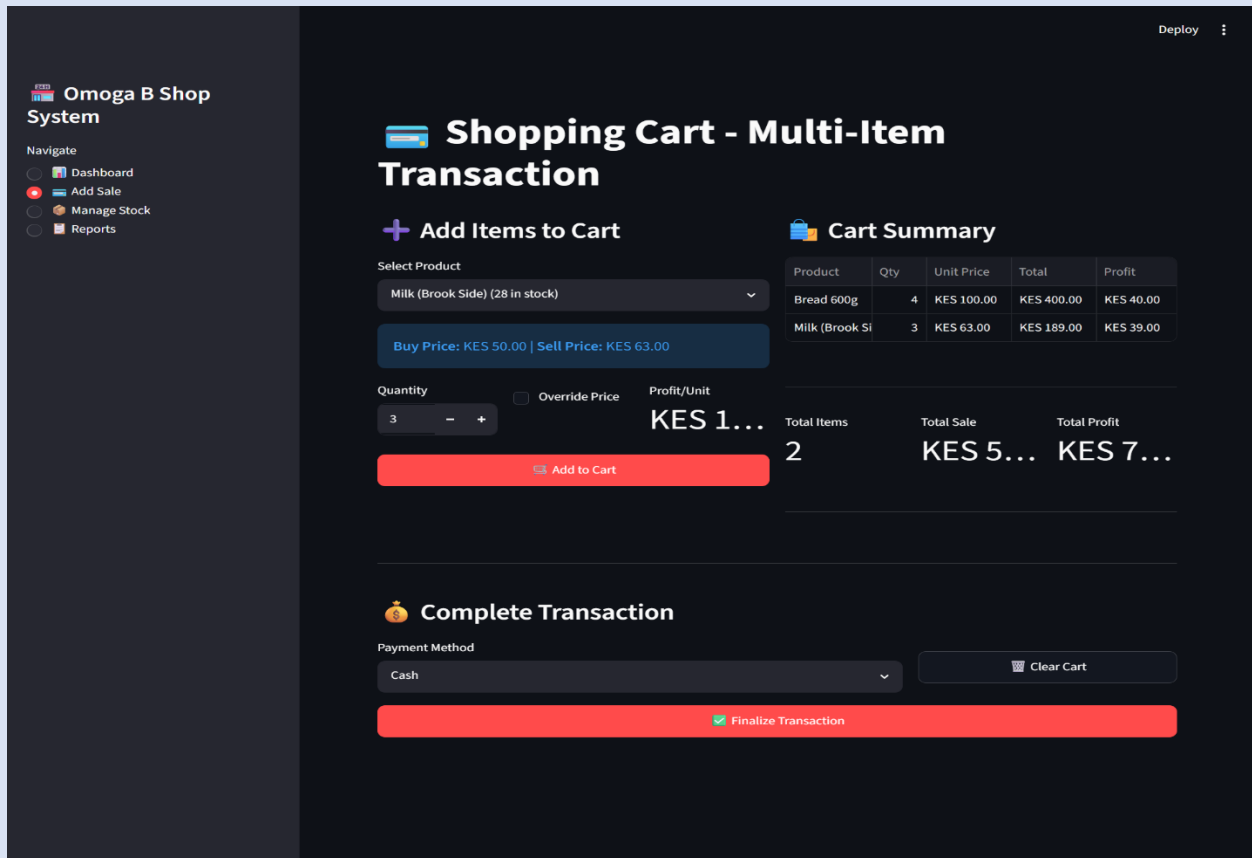
- A **decision-making hub**
- A **performance monitoring tool**
- A **real-time business snapshot**

It reduces the need to navigate multiple pages by consolidating sales, profit, and inventory insights into one view.



B. Add Sales Tab - Overview & Functionality

The **Add Sales** tab serves as the point-of-sale (POS) interface, enabling users (shopkeeper) to process customer purchases efficiently through a multi-item shopping cart system. It is designed to support the sale of multiple products within a single transaction, improving speed, accuracy, and customer service.



Product	Qty	Unit Price	Total	Profit
Bread 600g	4	KES 100.00	KES 400.00	KES 40.00
Milk (Brook Si	3	KES 63.00	KES 189.00	KES 39.00

Total Items	Total Sale	Total Profit
2	KES 5...	KES 7...

Key Functions in the Add Sales Tab

1. Add Items to Cart

This section allows users(shopkeeper) to build a customer's order before checkout.

Features include:

- Product selection from available inventory
- Display of current stock levels
- Visibility of buying and selling prices
- Quantity adjustment using increment and decrement controls
- Optional price override functionality
- Automatic calculation of profit per unit
- Add selected products to the shopping cart



Purpose: Enables quick and accurate entry of customer purchases.

2. Cart Summary

Displays all products added to the current transaction.

Information shown includes:

- Product name
- Quantity
- Unit price
- Total amount per item
- Profit generated per item

The system also automatically calculates:

- Total items in the cart
- Total sales value
- Total profit for the transaction

Purpose: Provides a real-time overview of the entire sale before checkout.

3. Complete Transaction

This section finalizes the sale.

Features include:

- Payment method selection (e.g., Cash, M-Pesa, Card, Credit)
- Transaction confirmation and processing
- Automatic sales recording
- Inventory update based on quantities sold
- Profit calculation and storage

Purpose: Ensures all sales are accurately recorded and reflected in inventory and financial reports.

4. Clear Cart

Allows users to remove all items from the current shopping cart and start a new transaction.

Purpose: Helps correct mistakes or cancel incomplete transactions before checkout.

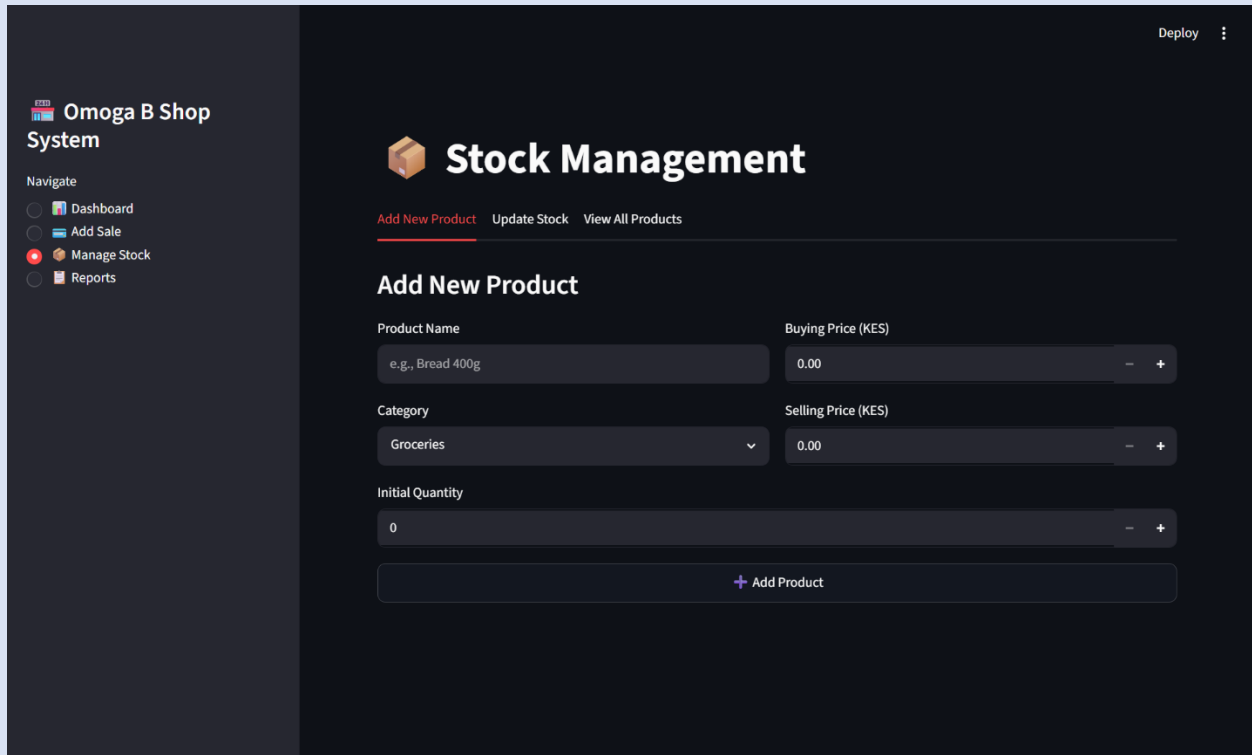
Overall UI Role of the Add Sales Tab

The Add Sales tab functions as the system's **sales processing center**, streamlining the entire sales workflow from product selection to payment completion. It enhances operational efficiency by supporting multi-item transactions, real-time calculations, and automatic inventory updates within a single interface.



C. Manage Stock Module

The **Manage Stock** tab serves as the inventory management center of the Retail Sales & Profit Management System. It enables users to add new products, update existing inventory records, and monitor stock levels in real time.



Key Features Displayed

1. Add New Product

This section allows users to register new products into the inventory database.

Input Fields:

- **Product Name** – Enter the name of the item (e.g., Bread 400g).
- **Category** – Select the product category from a dropdown list.
- **Buying Price (KES)** – Cost price paid to suppliers.
- **Selling Price (KES)** – Retail selling price charged to customers.
- **Initial Quantity** – Opening stock quantity available for sale.

Function:

- Creates a new inventory record.
- Stores product details in the SQLite database.
- Makes the product immediately available for sales transactions.



2. Update Stock

Accessible through the **Update Stock** tab, this feature allows users to modify:

- Product category
- Buying price
- Selling price
- Available stock quantity

Function:

- Keeps inventory records accurate.
- Supports stock replenishment and price adjustments.
- Updates the product's last modified timestamp.

3. View All Products

The **View All Products** tab provides a complete inventory listing.

Information Displayed:

- Product ID
- Product Name
- Category
- Buying Price
- Selling Price
- Current Stock Quantity

Function:

- Enables inventory monitoring.
- Helps identify low-stock items.
- Provides a quick overview of all products in the shop.

4. Inventory Control

The module works closely with the Sales module.

When a sale is recorded:

1. Product quantity is automatically reduced.
2. Updated stock levels are reflected immediately.
3. Inventory data remains synchronized with sales transactions.



Business Value

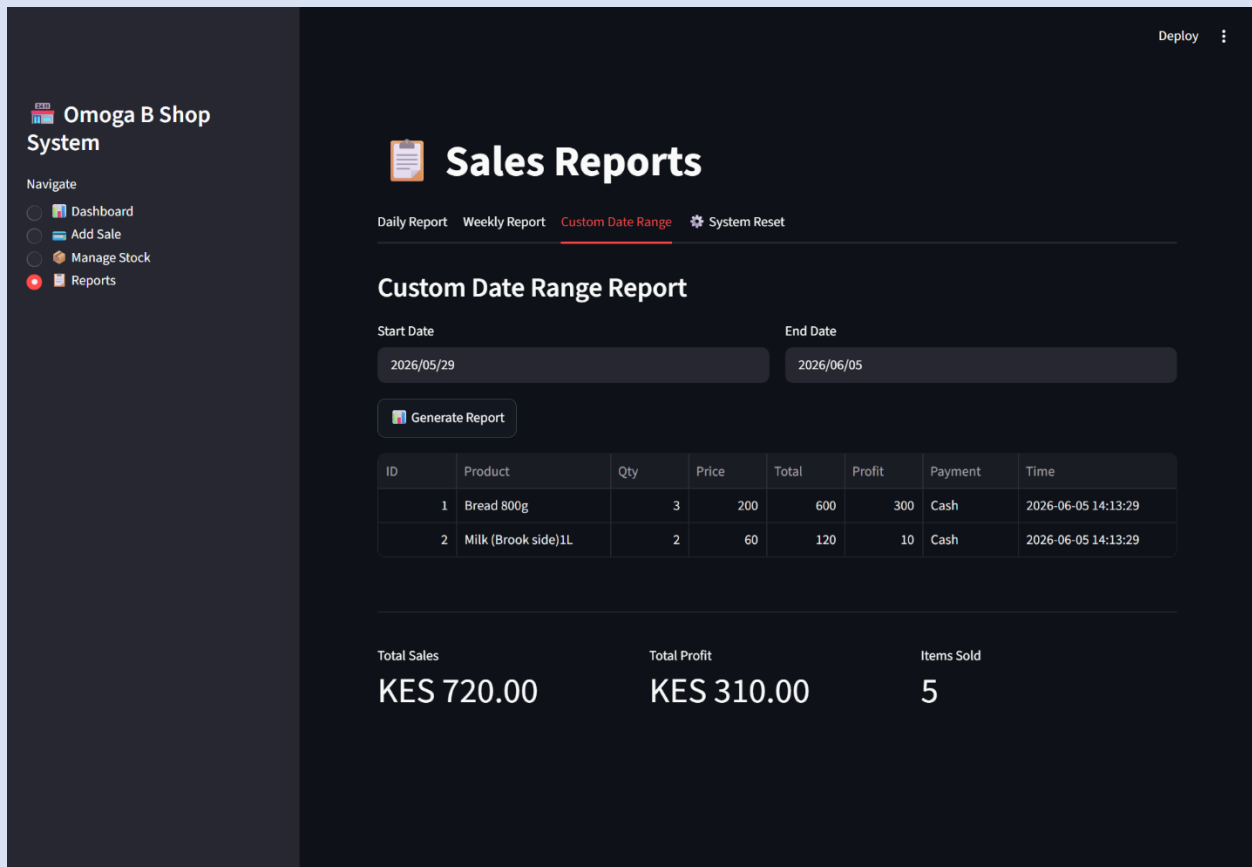
The Manage Stock module helps shop owners:

- Maintain accurate inventory records.
- Prevent stock discrepancies.
- Monitor product pricing.
- Track stock availability in real time.
- Support efficient purchasing and restocking decisions.

In the overall system architecture, this module acts as the **Inventory Management Engine**, ensuring that all products sold through the POS system are properly tracked and managed.

D. Reports Module

The **Reports** tab is the business intelligence and performance monitoring component of the Retail Sales & Profit Management System. It enables users to analyze sales performance, profitability, and transaction history over different time periods, providing valuable insights for informed decision-making



The screenshot displays the 'Omega B Shop System' interface. On the left is a dark sidebar with a 'Navigate' menu containing icons and labels for 'Dashboard', 'Add Sale', 'Manage Stock', and 'Reports' (which is highlighted with a red dot). The main content area has a dark background. At the top right of this area is a 'Deploy' button with a dropdown arrow. Below this is the 'Sales Reports' section, featuring a clipboard icon and the title 'Sales Reports'. Underneath are tabs for 'Daily Report', 'Weekly Report', and 'Custom Date Range' (which is selected and underlined), along with a 'System Reset' button. The 'Custom Date Range Report' section follows, with 'Start Date' (2026/05/29) and 'End Date' (2026/06/05) input fields, and a 'Generate Report' button. Below these is a table with 8 columns: ID, Product, Qty, Price, Total, Profit, Payment, and Time. It contains two rows of data. At the bottom, there are three summary cards: 'Total Sales' (KES 720.00), 'Total Profit' (KES 310.00), and 'Items Sold' (5).

ID	Product	Qty	Price	Total	Profit	Payment	Time
1	Bread 800g	3	200	600	300	Cash	2026-06-05 14:13:29
2	Milk (Brook side)1L	2	60	120	10	Cash	2026-06-05 14:13:29

Total Sales

KES 720.00

Total Profit

KES 310.00

Items Sold

5



Overview of the Interface

The Reports module contains four reporting sections:

- **Daily Report**
- **Weekly Report**
- **Custom Date Range Report**
- **System Reset**

In the screenshot, the **Custom Date Range Report** tab is selected.

1. Custom Date Range Report

This feature allows users to generate sales reports for any selected period.

Input Parameters

Start Date

- Defines the beginning of the reporting period.

End Date

- Defines the end of the reporting period.

In the example shown:

- Start Date: 2026/05/29
- End Date: 2026/06/05

The system retrieves all sales transactions recorded between these dates.

2. Generate Report Function

The **Generate Report** button executes a database query that retrieves all sales records within the selected date range.

Functionality:

- Filters transactions by date.
- Retrieves sales information from the SQLite database.
- Displays detailed transaction records.
- Calculates summary statistics automatically.



3. Transaction Details Table

The report displays all transactions matching the selected date range.

Information Displayed

Column	Description
ID	Transaction identifier
Product	Product sold
Qty	Quantity sold
Price	Unit selling price
Total	Total sales amount
Profit	Profit earned from transaction
Payment	Payment method used
Time	Date and time of transaction

Sample Transactions

Bread 800g

- Quantity Sold: 3
- Selling Price: KES 200
- Total Sales: KES 600
- Profit: KES 300

Milk (Brookside)

- Quantity Sold: 2
- Selling Price: KES 60
- Total Sales: KES 120
- Profit: KES 10

This detailed transaction view enables users to audit sales activities and verify revenue generation.



4. Summary Performance Metrics

At the bottom of the report, the system automatically calculates key performance indicators (KPIs).

Total Sales

KES 720.00

Represents total revenue generated during the selected period.

Total Profit

KES 310.00

Represents net profit earned from all recorded sales within the selected period.

Items Sold

5

Represents the total number of units sold across all transactions.

Additional Reporting Features

Daily Report

Provides a summary of:

- Daily sales revenue
- Daily profit
- Total items sold
- Number of transactions

Useful for end-of-day business reviews.



Weekly Report

Aggregates sales performance over the current week.

Provides:

- Weekly sales totals
- Weekly profit totals
- Daily performance breakdowns
- Sales trend monitoring

Useful for identifying high-performing days.

System Reset

Administrative maintenance section used to:

- Clear sales records
- Clear inventory records
- Reset the entire database

Typically used for testing or starting a new business cycle.

Business Value

The Reports module transforms raw sales data into actionable business insights by enabling shop owners to:

- Track revenue and profitability.
- Monitor sales performance over time.
- Analyze transaction history.
- Identify top-selling products.
- Support stock replenishment decisions.
- Evaluate business growth and operational efficiency.

Within the overall system architecture, the Reports module functions as the **decision-support and business analytics component**, helping users understand how the business is performing beyond day-to-day transaction processing.



E. Future Enhancements

- User authentication (cashier logins)
- Multi-user support with role-based access
- Barcode scanning for faster sales entry
- Inventory alerts for low stock
- Export reports to PDF/Excel
- Product images
- Customer profiles and loyalty tracking
- Voice-based sales entry
- Distributed system architecture to support multiple independent shops (multi-tenant cloud deployment where each shop operates in isolation but shares a centralized backend with secure data separation, enabling scalability, remote access, and centralized management across different locations)

This enhancement implies upgrading your system so that:

- Each shop becomes a **tenant (isolated workspace)**
- All shops use the same system but **cannot see each other's data**
- A central backend (cloud server/API) manages:
 - Authentication
 - Databases
 - Reporting
- Shops access the system via web/mobile anywhere
- This can later be scaled to 10, 100, or 1000+ shops

Simple technical direction to be implemented in the future development

- Backend: **FastAPI / Django / Node.js**
- Database: **PostgreSQL or MySQL (multi-tenant schema design)**
- Deployment: **AWS / Azure / GCP / Docker + Kubernetes**
- Architecture: **Microservices or modular monolith → then scale to microservices**